Core System Commands

- `sudo`: Run commands with elevated (root) privileges.
  - **Purpose**: Allows a non-root user to execute administrative tasks securely.
  - **Example**: `sudo apt update` (update package lists).
- `su`: Switch to another user account (including root).
  - **Purpose**: Log in as another user or root for administrative purposes.
  - **Example**: `su -` (switch to root and load its environment).
- `whoami`: Display the current username.
  - **Purpose**: Verify which user is logged in.
  - **Example**: `whoami`.

File and Directory Management

- `ls`: List files and directories.
  - **Purpose**: View contents of a directory.
  - **Examples**:
    - `ls -l`: Detailed list.
    - `ls -a`: Include hidden files.
- `cd`: Change directory.
  - **Purpose**: Navigate between directories.
  - **Example**: `cd /etc`.
- `pwd`: Print the current working directory.
  - **Purpose**: Check your location in the file system.
- `mkdir`: Create new directories.
  - **Purpose**: Organize files by creating folders.
  - **Example**: `mkdir new_folder`.
- `rm`: Remove files or directories.
  - **Purpose**: Delete unwanted files or directories.
  - **Example**: `rm -rf /path/to/directory` (forcefully remove a directory).
- `cp`: Copy files or directories.
  - **Purpose**: Duplicate files for backup or organization.
  - **Example**: `cp file1.txt file2.txt`.
- `mv`: Move or rename files or directories.
  - **Purpose**: Relocate or rename files.
  - **Example**: `mv old_name.txt new_name.txt`.
- `find`: Search for files and directories.
  - **Purpose**: Locate files matching specific criteria.
  - **Example**: `find / -name file.txt`.
- `touch`: Create empty files.
  - **Purpose**: Quickly create files.

- **Example**: `touch new_file.txt`.

System Information

- `uname`: Display system information.
    - **Purpose**: Get details about the kernel, system name, version, etc.
    - **Example**: `uname -a`.
- `hostname`: Show or set the hostname of the system.
    - **Purpose**: Identify the machine in a network.
    - **Example**: `hostnamectl set-hostname new-hostname`.
- `df`: Display disk space usage.
    - **Purpose**: Check available and used disk space.
    - **Example**: `df -h` (human-readable format).
- `du`: Show the size of files and directories.
    - **Purpose**: Analyze disk usage.
    - **Example**: `du -sh folder_name`.
- `uptime`: Show system uptime.
    - **Purpose**: Monitor how long the system has been running.
- `free`: Show memory usage.
    - **Purpose**: Check RAM usage and availability.
    - **Example**: `free -h`.

Process Management

- `top`: Display real-time processes and resource usage.
    - **Purpose**: Monitor system performance.
- `ps`: List active processes.
    - **Purpose**: View details about running processes.
    - **Example**: `ps aux`.
- `kill`: Terminate processes.
    - **Purpose**: Stop unresponsive programs.
    - **Example**: `kill 1234` (terminate process with ID 1234).

Networking

- `ping`: Test connectivity to another host.
    - **Purpose**: Troubleshoot network issues.
    - **Example**: `ping google.com`.
- `ifconfig` **/** `ip`: View or configure network interfaces.

    - **Purpose**: Diagnose or modify network settings.
    - **Examples**:
        - `ifconfig`: Show IP address details.
        - `ip addr`: Newer alternative to `ifconfig`.
- **`curl`**: Fetch data from a URL.
    - **Purpose**: Test APIs or download files.
    - **Example**: `curl http://example.com`.
- **`wget`**: Download files from the web.
    - **Purpose**: Automate file downloads.
    - **Example**: `wget http://example.com/file.zip`.
- **`ssh`**: Securely connect to remote servers.
    - **Purpose**: Manage remote systems.
    - **Example**: `ssh user@hostname`.

Package Management

- **`apt` (Debian/Ubuntu)**: Manage software packages.
    - **Purpose**: Install, update, or remove packages.
    - **Examples**:
        - `sudo apt update`: Update package lists.
        - `sudo apt install package_name`: Install a package.
- **`yum` / `dnf` (RHEL/Fedora)**: Package manager for Red Hat-based systems.
    - **Purpose**: Similar to `apt`.
    - **Example**: `sudo yum install package_name`.
- **`pip`**: Install Python packages.
    - **Purpose**: Manage Python dependencies.
    - **Example**: `pip install package_name`.

User Management

- **`adduser`**: Create a new user account.
    - **Purpose**: Add users to the system.
    - **Example**: `sudo adduser new_user`.
- **`passwd`**: Change user passwords.
    - **Purpose**: Secure user accounts.
    - **Example**: `passwd username`.
- **`usermod`**: Modify user accounts.
    - **Purpose**: Change user privileges or groups.
    - **Example**: `sudo usermod -aG group_name username`.
- **`who`**: Show who is logged in.
    - **Purpose**: Monitor active sessions.

File Viewing Commands

- `head`: View the first few lines of a file.
  - **Purpose**: Quickly check the beginning of a file.
  - **Example**: `head file.txt` (default: first 10 lines).
  - **Options**:
    - `head -n 5 file.txt`: View the first 5 lines of a file.
- `tail`: View the last few lines of a file.
  - **Purpose**: Monitor the end of a file, often used with log files.
  - **Example**: `tail file.txt` (default: last 10 lines).
  - **Options**:
    - `tail -n 20 file.txt`: View the last 20 lines.
    - `tail -f file.txt`: Continuously monitor changes in a file.
- `cat`: Concatenate and display the contents of a file.
  - **Purpose**: View the entire file content at once.
  - **Example**: `cat file.txt`.
- `more`: View file content one screen at a time.
  - **Purpose**: Helpful for long files.
  - **Example**: `more file.txt`.
- `less`: Similar to `more`, but with backward navigation.
  - **Purpose**: View large files efficiently.
  - **Example**: `less file.txt`.

Text Processing Commands

- `grep`: Search for patterns in text files.
  - **Purpose**: Locate specific strings or lines.
  - **Example**: `grep "pattern" file.txt`.
- `awk`: Process and analyze text files.
  - **Purpose**: Extract or format file content.
  - **Example**: `awk '{print $1}' file.txt`.
- `sed`: Stream editor for text transformation.
  - **Purpose**: Find and replace text in files.
  - **Example**: `sed 's/old/new/g' file.txt`.

Disk and File System Commands

- `df`: Show disk space usage.
  - **Purpose**: Monitor disk usage.
  - **Example**: `df -h` (human-readable format).
- `du`: Display the size of files and directories.

    - **Purpose**: Identify storage usage.
    - **Example**: `du -sh folder_name`.
- **`mount`**: Mount a file system.
    - **Purpose**: Access external drives or partitions.
    - **Example**: `sudo mount /dev/sdb1 /mnt`.
- **`umount`**: Unmount a file system.
    - **Purpose**: Safely detach drives or partitions.
    - **Example**: `sudo umount /mnt`.

Archiving and Compression

- **`tar`**: Create or extract archive files.
    - **Purpose**: Bundle files for storage or transfer.
    - **Example**:
        - `tar -cvf archive.tar file1 file2`: Create a tar archive.
        - `tar -xvf archive.tar`: Extract an archive.
- **`gzip`** / **`gunzip`**: Compress and decompress files.
    - **Purpose**: Save storage space or prepare for transfer.
    - **Example**:
        - `gzip file.txt`: Compress a file.
        - `gunzip file.txt.gz`: Decompress a file.
- **`zip`** / **`unzip`**: Compress and extract zip files.
    - **Purpose**: Compatible compression format for sharing.
    - **Example**:
        - `zip archive.zip file1 file2`: Create a zip file.
        - `unzip archive.zip`: Extract a zip file.